

COIT20246 Project

Project Specification

1 Summary

Design a network that meets the provided scenario requirements. Then, investigate options for using cloud services as part of the network. Finally, conduct a cybersecurity risk assessment for the scenario and recommend security controls.

2 Scenario

Truelec is an electrical contracting business that has been operating for over 20 years. The beginning of the business was humble, with just five electricians working on residential maintenance. However, via a strategic growth program, Truelec has expanded to undertake an impressive portfolio of industrial projects, including major building construction, gas projects, mining developments and telecommunications projects. The company has several branch offices located in Australian major cities such as Brisbane, Perth, Adelaide, and Hobart.

Let's assume the Truelec headquarters is in your city, with a central office that contains the main administrative and operational functions, including project management consultants, marketing, senior leadership teams, and ICT staff. If you are a student of the Melbourne campus, assume the Truelec headquarters is located in Melbourne. The total staff may vary between 50 and 75. Each branch office is equipped with a team of electricians, electrical engineers, site supervisors, and other support staff, with each branch office ranging in size from 15 to 30 staff.

In their last annual strategic planning session, the senior leadership team agreed to move to a new, larger headquarters and relocate several branch offices to new strategic locations closer to the main city. For the headquarters, they require a complete wired network for the new location, as well as a WiFi network. Truelec currently hosts many IT services internally, including websites, HR systems, accounting software, Customer Relationship Management (CRM) systems, as well as servers to ensure the availability and security of client data. The company also manages its own internal security system via a network of CCTV cameras, IoT sensors and RFID cards for access.

The company wants to design the network with redundancy in mind. The network design must ensure continued availability in case of device or link failure, especially for WANs and routers. The management of this company has approached you to upgrade the network infrastructure and to design the computer network for the headquarters, as well as to provide design and security advice for the entire organisation (including branches). Use the above information provided for your proposed design. Where assumptions are made, you will need to clearly document and explain only the assumptions that your design depends on. Keep your design simple.

The management wants to see the design and hear your inputs on a variety of issues (including cyber security and cloud services). Then they will decide whether to further engage your services for the network build, operation, and maintenance.

3 Groups and Project Planning

3.1 Group Formation

This is a group project consisting of two (2) students from the same class (or with the Unit Coordinators permission, three students, e.g., when an odd number of students are in the class). Students should form their own groups, but if not formed within a reasonable time, the Unit Coordinator may allocate students to a group.

3.2 Project Plan

Before the start of Week 5, you must:

1. Form your group and setup project GitHub repository via GitHub Classroom. Instructions are available on Moodle.
2. Agree upon communication frequency (e.g., “every Sunday and Wednesday”, “during tutorial and every 2nd Thursday”) and methods (e.g., Zoom, face-to-face, Teams).
3. Create a schedule of the weeks in which you plan to complete each task.

You must include the communications plan and schedule in your GitHub repository in the file plan.md.

3.3 Group Contribution and Participation

All group members should contribute to all tasks; however, a group member may contribute more than the other(s) on a specific task. Overall, the contributions of each group member should be about the same.

One measure of your contribution to the group is your commits on the group GitHub repository. The commits must be frequent (not just before submission) and by all members of the group. Normally, a student will make a commit after a contribution, e.g., write a paragraph, create a diagram. At a minimum, you must commit to GitHub once for each day you work on the project. For example, if you draft the project plan on a Tuesday, then create the network design diagram on Wednesday, and then on Friday make further edits and write a paragraph about the design, then at minimum your group must have 3 commit (one for Tuesday, Wednesday and Friday). However, you are encouraged to make commits more frequently, especially when work on many parts of the project on a single day. Each student that makes the contribution must make the commit; allocating a single student to make commits on behalf of other students is not allowed. In summary, each student should be making multiple commits per week.

3.4 Problems in Groups and Changing Groups

Generally, a change of groups is not allowed once the project starts. If a student is not contributing to the project (e.g., they are absent or not responding to communications), the other student must inform the Unit Coordinator as soon as possible.

If a group is split, resulting in a student working on their own (e.g., one student withdraws from the unit or difficulties in the groups result in the Unit Coordinator splitting the students), the tasks will not change. That is, a student on their own must complete the same tasks as a group of 2 students. The tasks are designed such that a student can complete them on their own within a reasonable time. Ideally, working in a group of 2 is an advantage and the overall project can be completed in less time. However, in practice, there is an overhead of collaborating in a group. Therefore, the tasks remain the same no matter the group size.

4 Tasks

4.1 Network Design

4.1.1 List the Assumptions

The scenario does not specify all details – you need to make assumptions. You need to list the following to clarify the assumptions you made:

1. Location (city) of the headquarters
2. Locations (cities) of the branches (must be at least 3)
3. Number of staff in the headquarters
4. Number of staff in one branch office (that you design the network for – see below).

4.1.2 Design the Network

Design the network for the headquarters and for one branch office. Your design should include:

1. One or more network diagrams (using draw.io/diagrams.net),
2. Explanation of key design decisions, e.g., why you chose to design as you did, (avoid selecting a design approach not covered in this unit that you cannot adequately justify/explain, e.g., VLAN),
3. Explain the WiFi design, including listing important settings (and their values),
4. Address allocations, e.g., IPv4 address ranges to be used,
5. List of recommended hardware, e.g., recommended minimum specifications for equipment, with links to websites with the specifications and prices (AUD).

4.1.3 IP Addressing Requirements

The IP address ranges to be used are chosen by you, however they must meet the following requirements:

- Only /16 or /24 network masks
- The 1st decimal value, A, in any IP address, A.B.C.D, must be the last two digits of one of the group members student ID.
- The 2nd (B), 3rd (C) and 4th (D) decimal value in an IP address, A.B.C.D, can be any value that is logically correct.

For example, if a group has two members with IDs 12345678 and 12234506 then all IP addresses used in your network design must start with 78 or 6, such as 78.1.2.3, 78.43.0.0/16, 6.19.123.56, 6.100.24.0/24. Depending on your network design, you can select to use any of the student IDs or use both. Private address, such as 192.168.x.y, must not be used in any part of the network.

4.2 Cloud Services

The organisation currently keeps all data and servers on-premise, with most at headquarters. However, they are considering shifting some parts to the cloud. You have been directed to explore, evaluate, and recommend if cloud services are better suited for their operations.

4.2.1 Cloud Server

Truelec has a special-purpose booking app that requires three servers in the headquarters and one server per branch office. Currently, all the servers are 5-year old Dell PowerEdge Tower Servers. The company is considering replacing the current servers with similar, new servers or using cloud virtual machines instead. You must:

1. Find the price (AUD) of one cloud VM from at least two different cloud providers, choosing from Azure, AWS, or Google Cloud. Use the official calculators from those providers (use the given

links), and you must export the price estimate from the calculator and upload the files to your GitHub repository. You must also recommend one cloud provider to the organisation. Give a reason for the selected cloud provider. You must summarise the values you chose for important specifications (e.g., region, operating system, CPU, storage) and justify your selection (e.g., why did you choose each value). The specifications should be the same or similar across cloud providers (e.g., do NOT choose a 1 TB disk for a backup server in AWS and then choose a 5 TB disk for a backup server in Azure; that is an unfair comparison).

2. Calculate the total cost of using all cloud VMs to replace the existing servers over a 5-year period. You must also discuss the advantages/disadvantages of using the cloud VMs vs buying new servers.

The reader of your report should not be required to perform analysis nor make necessary deductions on your behalf if your report is clear and complete.

4.3 Security

4.3.1 Conduct cyber security risk assessment

You are to conduct a mini cyber security risk assessment of your network. By “mini” we mean you only conduct a subset of steps of a typical full risk assessment (due to lack of time and information available to you). You must use the risk assessment template and follow the process provided in the unit. Your risk assessment must consider vulnerabilities across at least 8 of the 12 information security threats, must contain assets of each asset type, and there must be at least 4 different data assets considered.

4.3.2 Recommend security controls

Select the data asset rated with the highest risk (from the previous risk assessment). Consider the security controls covered in the unit (in the lecture or in NIST SP800-53) and select three of the most useful controls for your design. Recommend at least one specific way you can use each of these controls in the project scenario to reduce the risk of the selected data asset. You must explain how the control reduces the risk, give details of how the control can be used (e.g., referring to devices, data and users in the project scenario, recommend specific technologies or approaches), and should mention specific parts of your network design when relevant (e.g., where in the network will the control be implemented; what will need to change in the network). You should also discuss any disadvantages of introducing the controls from the perspective of the users.

4.4 Ethical and Social Issues

Identify specific data privacy and security concerns that the travel agency might face. Consider aspects such as the collection, storage, and use of customer data, as well as potential vulnerabilities in the network. You should explain: the type of data collected by the travel agency, potential risks associated with unauthorised access or data breaches; relevant data protection regulations and their implications for the company (e.g., state, national or international laws). Consider the potential impact (who is impacted? How are they impacted? How significant?) if there is a data breach. Explain and provide source references.

4.5 Project Reflection

You must reflect on the performance of your project, including your own contributions by:

1. Include a screenshot showing commits of all team members so far (similar to Task 4 of Tutorial Handout 7).
2. List the actual split of tasks, e.g., a table or list showing the tasks by student A and the tasks by student B.
3. Compare the number of git commits of each student with the tasks performed, and discuss how they are related (e.g., reasons why the number of commits may be different for students,

whether the number of commits is proportional to the actual contributions of each student).

4. Identify how many weeks during the term all students in the group made commits, and comment if you think it was sufficient.
5. Consider what worked well in a group, and what issues you encountered. Recommended techniques you (or others) should use in future group projects (e.g., in units next term) to ensure successful teamwork. For each technique, explain what problem the technique is likely to solve or how it will help teamwork. You must refer to specific activities and techniques you actually performed/used in the project.

5 Report and Presentation

5.1 Written Report

The report must be in Markdown on your group project repository. A template and repository will be provided to you when you register for the project (see Moodle for link/information). There will be template Markdown files to use, however images and other supporting files (e.g., spreadsheets, diagrams) must also be included in your repository and linked from the relevant Markdown files.

The report must be presented in a professional technical report (e.g., title, authors, section headings, images suitably sized) and should include sections for each of the tasks. You may use sub-sections where relevant. Include your own pictures and tables where relevant; do not use images from other sources. Do NOT change any names of files given with the template.

There is no word limit or page limit. Several of the tasks involve design and explanation, and therefore, the level of detail (and length of the report) may vary significantly across groups.

There is no recommended number of references. It is not expected you will need to read research papers to complete this project, however you may need to research information from standards, industry best practices and websites (e.g., technical documentation, specifications). When you do gain key knowledge from other sources, including AI chatbots (e.g., ChatGPT), you must acknowledge that source (by referencing). Relevance and consistency (in format) of references are much more important than the number of references or following a specific format.

5.2 Recorded Presentation

The presentation will be a recorded video that must cover the following:

1. Explain briefly the key ethical issues that the organisation must consider, with each student explaining one issue. (Section 4.4)
2. A summary of who contributed to the different tasks during the project, and how many Git commits each student made. (Section 4.5)
3. A summary of how your group worked together, e.g., how often you met, where did you meet (in- class, after class, online), what communication mechanisms you used. (Section 4.5)
4. Reflection on what went wrong and how you can improve in the future. (Section

The presentation must meet the following requirements:

- All students in the group must speak, e.g., take in turns covering the different parts.
- On the first appearance, each speaker must identify themselves before presenting. This includes showing the speaker's face on the screen (either full screen or in thumbnail).
- No PowerPoint slides are needed. You are recommended to speak without notes. You may show parts of your report on the screen if relevant but should not be reading your report.

- Duration of approximately 5 minutes (about 1 minute for each of the items). The duration must be more than 4 minutes and less than 7 minutes.
- The presentation should be submitted as a single file. The file format must be mp4.

6 Submissions

There are two submissions for the Project:

1. Project Part 1: must contain the completed Network Design (Section 4.1). The other tasks/sections are NOT needed in Project Part 1 (and will not be marked).
2. Project Part 2: must contain all remaining tasks. The Network Design will NOT be re-marked in the Project Part 2.

6.1 What to include in your submission?

In your group GitHub project repository, you must include:

- Multiple Markdown files, following the template provided to you,
- The completed README.md file,
- One or more diagrams.net (.drawio) diagram files and corresponding exported PNG/JPG images, with those images linked and displayed in the relevant .md file (e.g., network design images should be in network.md),
- One risk assessment spreadsheet (.xlsx),
- One (or more) PNG/JPG screenshot of the TVAMatrix sheet from the risk assessment, linked from your security.md file. If the sheet is too large, take multiple screenshots,
- Two (or more) exports of cloud service price estimates. These are usually a spreadsheet, e.g., xlsx or csv, and should be linked from the cloud.md file,
- Any other supporting files linked from your Markdown files. These other files must be in subdirectories of the project repository.

6.2 How many files do you submit?

In Moodle you must submit:

Project Part 1 - total of 4 files:

- A PDF of the corresponding markdown pages from your GitHub repository: README.pdf, plan.pdf, network.pdf. (i.e. three PDF files)
- A ZIP of the entire GitHub repository.

Project Part 2 – total of 5 files and one video via Echo360:

- A PDF of the corresponding markdown pages from your GitHub repository: cloud.pdf, security.pdf, ethics.pdf, reflection.pdf. (i.e. four PDF files)
- A ZIP of the entire GitHub repository.
- A MP4 for the recorded presentation, uploaded to Moodle via Echo360.

6.3 Submitting PDF of Markdown

The PDF files must be created by viewing the corresponding Markdown (.md) file in your web browser and then selecting Print, and Save as PDF. You must have one PDF for each Markdown file, e.g., network.md produces network.pdf.

6.4 How do you create a video?

You are recommended to use Zoom to create the video for Project Part 2. All students in the group should

have their webcam on, and one (or multiple) student(s) share their screen. Use the “Record to your computer” option in Zoom. Watch your recording before submission, ensuring it is correct, e.g. contains audio, student faces are shown, the screen is shown.

6.5 Submitting MP4 video in Echo360

To submit the video, you must use Echo360. Instructions are at:

<https://moodle.cqu.edu.au/mod/page/view.php?id=3340315&forceview=1>

It is important the video is submitted using these instructions with Echo360. Do NOT include your MP4 in the ZIP.

6.6 How to submit in a group

Moodle submission is configured so that you must be a member of a group to submit. The first member of the group uploads the file for submission. Each other member of the group must then confirm the submission in Moodle. The submission is not finalised until all group members have confirmed the submission. A submission left in ‘draft’ is not finalised.

6.7 What is the submission deadline?

The deadline for the submission is given on Moodle. The time in at which your group finalises the submission of the files on Moodle is taken as the submission time. Late penalties of 5% per part of the day late are calculated based on the Moodle submission time.

6.8 What is marked?

The marker may mark either the version on GitHub, the ZIP on Moodle or the PDF on Moodle – they all should be identical. If they are different, then the marker may give a score based on the worst of the three. It is important once you submit on Moodle the GitHub repository does not change. Any changes to the GitHub repository after Moodle submission will not be marked and may be considered as academic misconduct.

6.9 Is plagiarism detection (e.g. Turnitin) used?

Yes.

Your submission will be automatically passed to the CQU Turnitin service for plagiarism detection. In addition, our unit specific plagiarism software may be used as well (e.g. that detects similarities across images, drawings and packet captures, not just text). It is important you do all work yourself, not re-using work from previous terms or other groups.

Turnitin may provide a tentative similarity score for your submission. The teaching team will not normally respond to questions from students about their similarity score (e.g. “why is my similarity score so high?”, “can I re-submit to bring my similarity score down?”). We will use the similarity score and other measures (including our unit specific software results and professional judgement by markers) to determine if there is potential academic misconduct.

7 Marking Criteria

Part 1 has a total of 12 marks.

Part 2 has a total of 18 marks.

You can find a detailed marking rubric on Moodle.

7.1 Project Professionalism

You are required to apply industry best practices and professional behaviour in this project. This includes:

- Planning your project, including allocating tasks to each group member and agreeing on a communication plan for the project.
- Using a Git repository, with regular commits by all group members.
- Following requirements (e.g., this Project Specification), including submitting files in the required format.
- Creating data specific and unique to your group.
- Communicating in a professional manner in both a written report and a presentation.

If you do not demonstrate project professionalism (in terms of, for example, using your Git repository or submitting the correct files), then your mark will be limited to a set percent of the total available marks, that is you will not be able to receive full marks, and may limited to less than 50% in the project (a fail score). The detailed marking rubric on Moodle defines the limits based on project professionalism.

Submissions in the wrong format (e.g., GitHub not used) may not be marked (i.e., 0 marks and no feedback).

7.2 Individual Marks

While typically all members of a group will receive the same score, in some cases different marks may be awarded. This will be based on individual student contributions, e.g., measured by participation in tutorials, commits to GitHub, stated contributions, interviews with students. If a student does not make any reasonable contributions via GitHub, they may be awarded 0 marks.